

Artin 3

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Problem 4.9.

Proof. Use the trivial character and χ_2 to find out size of the last conjugacy class. Let $|[y]| = a$ and $\chi_2(y) = b$. By orthogonality relations, $1(1) + 2(1) + 2(1) - 3 + ab = 0 \implies ab = 3$. $a \in \mathbb{N}$, 3 only has integer factors 1 and 3. If $a = 1$, then $|Z(G)| = 3$ but $|G| = 10$ and $3 \nmid 10$, thus $a = 3$. Thus $b = 1$. Fill in the rest using row orthogonality relations and the regular representation

	(1)	(1)	(2)	(2)	(3)	(3)
	1	u	v	w	x	y
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	-1	1
χ_3	1	-1	1	-1	i	-i
χ_4	1	-1	1	-1	-i	i
χ_5	2	2	-1	-1	0	0
χ_6	2	-2	1	1	0	0
χ_{reg}	12	0	0	0	0	0

y must be x^{-1} because x^{-1} isn't anywhere else otherwise there would have been another column with i .

$u \in Z(G)$, thus $\langle u \rangle \in Z(G)$, $|Z(G)| = 2 \implies |u| = 2$.

G has a sylow 3 and a sylow 2 subgroup. The number of sylow 3 subgroups has to divide 2 and be 1 mod 3, thus there is only 1. Thus the sylow 3 subgroup is normal. Thus $\mu_3 \triangleleft G$.

Since $\forall$ 1 dimensional representations, $\chi(v) = 1$, $v \in [G, G]$. Only 3 elements are in $[G, G]$, elements of $[v]$ and 1. Thus $[G, G] = \mu_3$. Thus $|v| = 3$.

$|Z(w)| = 6$. $\mu_2 \leq Z(w)$ because $u \in Z(w)$. w cannot have order 3 because there is only 1 order 3 subgroup of G , $[G, G]$ by sylow theorems. $|w|$ cannot be 2, because its diagonalizable and thus its eigenvalues are 1 and / or -1. There is no possible sum of 2 numbers 1 and / or -1 which gives you 1. Thus $|w| = 2$, then having $\chi_6(w) = 1$ would not be possible. Thus $|w| = 6$.

$|Z(x)| = 4$. Since $\chi_3(x) = i$, $|x| = 4$. Thus $|x| = 4$. Similarly, $|y| = 4$, because $\chi_4(y) = i$, $|y| = 4$ and $Z(y) = 4$.

$Z(w)$ is normal because $|Z(w)| = 6$ and index 2 subgroups are normal. D_3 is not a normal subgroup in G because then either x or y is in D_3 because you need 1 conjugacy class of order 3 but then the conjugacy class would be missing x^{-1} or y^{-1} because $x^{-1} = y$.

μ_3 is normal as discussed above.

$Z(G) \cong \mu_2$ is normal because the center is always normal in a group.

If $\mu_2 \times \mu_2$ or μ_4 were normal, then there is a isomorphism from $G \rightarrow \mu_3 < C^*$, however in the one dimensional representations there are no characters which are 3rd roots of unity except for 1.

The only non abelian groups of order 12 are D_6 , $\mu_3 \rtimes \mu_4$, and A_4 . A_4 isn't possible because there are no elements of order 6. D_3 is normal in D_6 but $D^3 \not\triangleleft G$, thus $G \not\cong D_6$. Thus $G \cong \mu_3 \rtimes \mu_4$. $\square$

Problem 4.10.

Proof. You are missing a row of the character table, use the regular representation to find the final row.

	(1)	(4)	(5)	(5)	(5)
	1	a	b	c	d
χ_1	1	1	1	1	1
χ_2	1	1	-1	-1	1
χ_3	1	1	$-i$	i	-1
χ_4	1	1	i	$-i$	-1
χ_5	4	-1	0	0	0
χ_{reg}	14	0	0	0	0

$a \in Z(a) \cong \mu_5$. Thus $|a| = 5$.

Since $\chi_4(b) = i$ and $\chi_3(c) = i$, $4||b|$ and $4||c|$. $Z(b) = 4$ and $Z(c) = 4$, thus $|b| = 4$ and $|c| = 4$.

$Z(d) = 4$. Thus $|d| = 2, 4$. $|\ker \chi_2| = 10$, thus $|d| \mid 10 \implies |d| = 2$.

$|\ker \chi_2| = 10$ and $\ker \chi_2 = N \triangleleft G$. Assume the contrary $N = \mu_{10}$, $\exists x \in \mathbb{N}$ such that $|x| = 10$ $x \in Z(x)$ but the only choices for $|Z(x)| = \{5, 4\}$ but $\langle x \rangle \leq Z(x)$. Thus by contradiction, $N \neq \mu_{10}$. Thus $N = D_5$

$D_5 \triangleleft G$, $C_5 = \ker \chi_4 \triangleleft G$. Since normal subgroups are disjoint unions of conjugacy classes, and no disjoint union produces a set of 2 and 4 elements, $\mu_2, \mu_4 \ntriangleleft G$. $\square$

Problem 4.11.

Proof. $\ker \chi_4 = \{e\} \cup [a] \triangleleft G$. $\ker \chi_4 \cong C_7$ because only group with 7 elements is C_7 . Furthermore $N = \ker \chi_3 = \{e\} \cup [a] \cup [b] \triangleleft G$. Since $a \in C_7$ and $a \neq 1$ then $|a| = 7$. Moreover since $\chi_4(b) = -1$, $2||b|$. However since, $|b| \mid 14$, it means $|b| = 2, 14$. However if $|b| = 14$, N would be μ_{14} . $b \in Z(b)$. However, $|Z(b)| = 6$. Thus by contradiction, $|b| \neq 14$. Thus $|b| = 2$ and $\langle b \rangle = \mu_2$. Furthermore since $\ker \chi_4 \triangleleft N$ because it is also the kernel of the character restricted to N , we know $\mu_7 \triangleleft N$. We know, N is not isomorphic to $\mu_7 \times \mu_2$ because that is μ_{14} . The only other group is D_{14} . This is true because there are only 2 choices for the 2 sylow group of N , its either normal or there are 7 conjugate subgroups. If it was normal, then it would be $N \cong \mu_{14}$. Thus $\mu_2 \rightarrow \text{Aut}(\mu_7) \cong \mu_6$, has only 1 nontrivial possibility $x \rightarrow x^3$. Thus $N \cong D_7$.

Character Table of D_7 . For the one dimensional representations, $y^2 = e \implies y = \pm 1$. Furthermore since $x^7 = 1$ and $xyx^{-1} = x^{-1}$ and C^* is abelian, $x = x^{-1} \implies x^2 = 1$, thus $x = 1$

From the irreducible representations of D_n homework, we know the 2 dimension irreducibles of D_7 have the following form. $x \rightarrow \begin{bmatrix} \rho^k & \\ & \rho^{-k} \end{bmatrix}$ where $\rho = e^{2\pi i/7}$ (8th root of unity) and $k \in \{1, 2, 3\}$ and

$y \rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. In any of the 3 2d representations, the trace of y is 0. $\text{tr}(x) = \rho^k + \rho^{-k} = 2 \cos(2k\frac{\pi}{7})$

	e	x	x^2	x^3	y
χ_A	1	1	1	1	1
χ_B	1	1	1	1	-1
χ_C	2	$2 \cos(2\frac{\pi}{7})$	$2 \cos(4\frac{\pi}{7})$	$2 \cos(6\frac{\pi}{7})$	0
χ_D	2	$2 \cos(4\frac{\pi}{7})$	$2 \cos(6\frac{\pi}{7})$	$2 \cos(2\frac{\pi}{7})$	0
χ_E	2	$2 \cos(6\frac{\pi}{7})$	$2 \cos(2\frac{\pi}{7})$	$2 \cos(4\frac{\pi}{7})$	0
χ_{reg}	14	0	0	0	0

Thus $\chi_1|N = \chi_2|N = \chi_3|N = \chi_A$, $\chi_4|N = \chi_5|N = \chi_6|N = \chi_B$. $\chi_7|N = \chi_C + \chi_D + \chi_E$.

$|G| = 42 = 2 * 21 = 3 * 14 = 7 * 2$. The number of 2 sylow subgroups of G , has to divide 21 and be 1 mod 2. There are 3 such possibilities, 3, 7, 21, however it cannot be 3, because we know N has 7 μ_2 subgroups which are conjugate within itself. Furthermore since all the 2-sylow subgroups are conjugate and 7 of them are part of a normal subgroup, they all must be. Assume the contrary A is a 2-sylow subgroup not in N and B is one of the 7 2-sylow subgroups in N . Thus $\exists g \in G \ni gAg^{-1} = B \implies A = (g^{-1})B(g^{-1})^{-1}$, but since N is normal $(g^{-1})B(g^{-1})^{-1} \leq N$ so $A \leq N$. Thus by contradiction, $\nexists$ a 2 sylow subgroup of G which is not in N . Thus there are 7 2 sylows.

The number of 3 sylow subgroups of G has to divide 14 and be 1 mod 3, there is only 1 possibility for this, 7.

The number of 7 sylow subgroups of G has to divide 2 and be 1 mod 7, there is only 1 possibility for this 1, which is true because $\ker_4 \cong C_7$ was normal and thus only 1 7 sylow subgroup existed.

$|Z(c)| = 6$, and since $\chi_2(c) = \omega$ and $\chi_6(c) = -1$ where ω is a 3rd root of unity implies, $3||c|$ and $2||c|$. Thus $|c| = 6$. Similarly, $|d| = 6$ since $\chi_3(d) = \omega$ and $\chi_6(d) = -1$.

Finally there are 7 conjugate μ_3 subgroups so there have to be at 2 conjugacy classes with at least 7 elements of order 3 or at 1 conjugacy class with at least 13 elements, the latter doesn't exist. $[c], [d]$ do not qualify for the first one because there are 6 elements for each conjugacy which could have orders of 7 because $|c| = 6$ and $|d| = 6$. The only two remaining conjugacy classes with at least 7 elements are $[e]$ and $[f]$. Since $\chi_2(e) = \omega$ and $\chi_2(f) = \bar{\omega}$, it means $3||e|$ and $3||f|$. e and f must have orders of 3, by the fact above because there need to be 2 conjugacy classes with at least 7 elements with order of 3. $[c]$ and $[d]$ do not qualify so $[e]$ and $[f]$ remain which only have 7 elements and thus all elements must have orders of 3. Thus $|e| = 3$ and $|f| = 3$.

A subgroup of order 21 is normal because it would have index 2. No subgroup of order 21 is abelian because $[1]$ must be in it and thus so must $[a]$, but $[a]$ has non 1 dimensional irreducibles and abelian groups do not have any multidimensional irreducibles. There is only 1 nonabelian subgroup of order 21, $\langle x, y | x^7 = y^3 = 1, y^{-1}xy = x^2 \rangle$.

D_7 is normal as shown above and C_7 is normal as shown above.

C_{14} cannot be normal because if it was there would exist a map from $G \rightarrow \mu_3 \leq C^*$ which would be 1 dimensional representations. However any 1 dimensional representation which maps $G \rightarrow \mu_3$ has a kernel of D_7 as seen above. Thus since a map doesn't exist like so, C_{14} cannot be normal.

C_6 cannot be normal, because normal subgroups are disjoint unions of conjugacy classes, and no disjoint union of conjugacy classes which include 1 give a set of size 6.

μ_3 and μ_2 are not normal because there exist 7 2 and 3 sylows each.

Thus the only normal subgroups are $\langle x, y | x^7 = y^3 = 1, y^{-1}xy = x^2 \rangle, D_{14}$, and μ_7 . $\square$

Problem 4.12.

Proof. 1. $\sigma' = \sigma \circ C_a$ where $a \in G$. $C_a \in \text{Aut}_{\text{group}}(H)$ because H is normal and σ is a homomorphism from $G \rightarrow GL(V)$. $\forall g, h \in H$, $\sigma'(gh) = \sigma(gha^{-1}) = \sigma(aga^{-1}aha^{-1}) = \sigma(aga^{-1})\sigma(aha^{-1}) = \sigma'(g)\sigma(h)$. Thus σ' is a homomorphism from $G \rightarrow GL(V)$ thus σ' is a representation.

2. $\chi = \text{tr} \circ \sigma$ and $\chi' = \text{tr} \circ \sigma'$. $\forall h \in H$, $\chi(h) = \text{tr}(\sigma(h))$ and $\chi'(h) = \text{tr}(\sigma(aha^{-1})) = \text{tr}(\sigma(a)\sigma(h)\sigma(a)^{-1}) = \text{tr}(\sigma(h)) = \chi(h)$ since trace is conjugacy invariant. By Frobenius's Theorem, since the characters are equal and thus $\sigma \cong \sigma'$.

3. Since $[G : H] = 2$, $\forall a, b \notin H$, $a, b \in aH \implies b = ah'$ for some $h' \in H$. Thus $\forall h \in H$, $\sigma''(h) = \sigma(bhb^{-1}) = \sigma(ah'hb^{-1}a^{-1}) = \sigma(ah'a^{-1}aha^{-1}ah'^{-1}a^{-1}) = \sigma'(h')\sigma(h)\sigma(h')^{-1}$ since trace is conjugacy invariant, $\text{tr}(\sigma''(h)) = \text{tr}(\sigma'(h')\sigma(h)\sigma(h')^{-1}) = \text{tr}(\sigma'(h))$. By Frobenius's theorem, since the characters are equal, the representations are isomorphic.

□

Problem 5.2.

Proof. For $n \geq 5$, the only proper normal subgroup of S_n is A_n and since A_n is a index 2 subgroup of S_n , $S_n/A_n \cong \mu_2$. The map from $S_n \rightarrow S_n/A_n$ is the sign homomorphism. The only other normal subgroups are the trivial subgroup and the whole group.

For $n = 4$, the only proper normal subgroups of S_4 are A_4 and K_4 . S_4/K_4 is not abelian and thus cannot be isomorphic to a 1 dimensional representation which is isomorphic. $S_4/A_4 \cong \mu_2 \leq C^*$ and the map from $S_4 \rightarrow \mu_2$ is canonically the sign map.

For $n = 3$, the only proper normal subgroup is $\mu_3 \cong A_3$ which is index 2. The map $S_3 \rightarrow S_3/\mu_3 \cong \mu_2$ is canonically the sign map.

For $n = 2$, $S_2 \cong \mu_2$ and $n = 2$ $S_1 \cong \{e\}$.

$\forall n \in \mathbb{N}$, $S_n \rightarrow S_n$, isn't a 1 dimensional representation because $S_n \not\leq C^*$. The whole group sends $S_n \rightarrow e$ sends everything to 1 which is a 1 dimensional representation.

Thus the only 1 dimensional representations of S_n are the sign representation $S_n \rightarrow S_n/A_n \cong \mu_2$ and the trivial representation. □

Problem 5.3.

Proof. Suppose G is a group with exactly 2 irreducible characters of dimension 1. Let χ be the nontrivial 1 dimensional character. $G \leq S_n$ for some $n \in \mathbb{N}$. Thus there exists a natural injective homomorphism from $t : G \rightarrow S_n$. Let ρ be the induced 1 dimensional character of S_n such that, $\chi = \rho \circ t$, ρ cannot be trivial because χ is nontrivial and $\forall g \in G, g \in S_n$. Since ρ is nontrivial, by 5.2 $\rho = \text{sign}$ thus $\forall g \in G, \chi(g) = \rho(t(g)) = \pm 1$, by the previous problem. Thus $\forall g \in G, \chi(g) = \pm 1$ □

Problem 5.4.

Proof. $\forall g \in G$, g is diagonalizable because $g^{|g|} = 1$ and thus satisfies the equation, $x^{|g|} - 1 = 0$. Thus g has eigenvalues, furthermore since all eigenvalues of g also satisfy the equation, all eigenvalues of g are roots of unity. Furthermore since $\chi(g) = \sum_{i=1}^d \lambda_i$ where λ_i are the eigenvalues of g . Thus, by the triangle inequality $|\chi(g)| = |\sum_{i=1}^d \lambda_i| \leq \sum_{i=1}^d |\lambda_i| = \sum_{i=1}^d 1 = d$. Furthermore equality only exists, if $\forall i \in \mathbb{N}_d$ all the λ_i are parallel and point in the same direction, and since they all have modulus 1, this means $\lambda_i = \lambda_j \forall i, j$. Thus $g = \lambda I$.

However if $\chi(g) = d$, by the above, $g = \lambda I$. Since the trace is the sum of the diagonal entries, $\sum_{n=1}^d \lambda = d\lambda = d \implies \lambda = 1$. Thus $g = I$. □

Problem 5.5.

Proof. $\widehat{G}$ is the group of 1 dimensional characters of G with the following multiplication rule, $\forall \alpha, \beta \in \widehat{G}$, $\forall g \in G, (\alpha\beta)(g) = \alpha(g)\beta(g)$.

1. **Closure:** $\forall \alpha, \beta \in \widehat{G}$ $\alpha\beta$ by the properties of the multiplication is also a map from $G \rightarrow C^*$. Furthermore $\forall g, h \in G$, $(\alpha\beta)(gh) = \alpha(gh)\beta(gh) = \alpha(g)\alpha(h)\beta(g)\beta(h)$. Since C^* is commutative, $(\alpha\beta)(gh) = \alpha(g)\alpha(h)\beta(g)\beta(h) = \alpha(g)\beta(g)\alpha(h)\beta(h) = (\alpha\beta)(g)(\alpha\beta)(h)$. Thus $\alpha\beta$ is a homomorphism from $G \rightarrow C^*$ and is the a 1 dimensional representation and $\widehat{G}$ is closed under multiplication.
2. **identity:** $\exists t \in \widehat{G}$, or the trivial map. $\forall \alpha \in \widehat{G}$, $\forall g \in G$, $(t\alpha)(g) = t(g)\alpha(g) = \alpha(g) = \alpha(g)t(g) = (\alpha t)(g)$. Thus t is the identity element.
3. **Inverse:** $\forall \alpha \in \widehat{G}$, let $\beta : G \rightarrow C^* \ni g \rightarrow \alpha(g)^{-1}$. $\forall g, h \in G$, $\beta(gh) = \alpha(gh)^{-1} = \alpha(h)^{-1}\alpha(g)^{-1} = \alpha(g)^{-1}\alpha(h)^{-1} = \beta(g)\beta(h)$, since C^* is commutative. Thus β is a homomorphism and a 1 dimensional representation. Thus $\beta \in \widehat{G}$. $\forall g \in G$, $(\alpha\beta)(g) = \alpha(g)\alpha(g)^{-1} = 1 = \alpha(g)^{-1}\alpha(g) = (\beta\alpha)(g)$. Thus α has inverse, β .
4. **Associativity:** $\forall \alpha, \beta, \gamma \in \widehat{G}$, $\forall g \in G$, $((\alpha\beta)\gamma)(g) = (\alpha\beta)(g)\gamma(g) = \alpha(g)\beta(g)\gamma(g) = \alpha(g)(\beta\gamma)(g) = (\alpha(\beta\gamma))(g)$, since multiplication is associative in $\mathbb{C}^*$. Thus $\widehat{G}$ is associative.

Thus $\widehat{G}$ is a group.

By the previous homework, $\widehat{G}$ is in bijection with $G/[G, G]$.

Suppose G is abelian, thus $G/[G, G]$ is the whole group because $[G, G] = e$. Thus $\widehat{G}$ is in bijection with G . Thus $|\widehat{G}| = |G|$

By Structure theorem all Finite abelian groups (Keel said we couldn't do this without structure theorem) are direct product of cyclic groups thus $G \cong C_{n_1} \times \dots \times C_{n_m}$. Furthermore $\forall m \in \mathbb{N}$ and $\forall n_1, \dots, n_m \in \mathbb{N}$, $Hom_{gr}(C_{n_1} \times C_{n_2} \dots \times C_{n_m}, \mathbb{C}) = \{x_1, x_2, \dots, x_m : x_1^{n_1} = x_2^{n_2} = \dots = x_m^{n_m}\} = C_{n_1} \times C_{n_2} \dots \times C_{n_m}$. Thus, any finite abelian group is isomorphic to its character. $\square$

Problem 5.6.

Proof. Suppose $G = C_n$. For any irreducible representation, $\rho : C_n \rightarrow C^*$, x gets sent to something which satisfies the equation $x^n = 1$. The only elements of C^* which satisfy this equation are elements of μ_n . Thus ρ is actually a map from $C_n \rightarrow \mu_n$. x generates C_n and ω generates μ_n . x can be mapped to any element $g \in \mu_n \ni g^n = 1$, $g = \omega^k$ for some $k \in \mathbb{N}_n$. Thus $\omega^{kn} = 1$. But since every element of μ_n satisfies the equation, $p(a) = a^n - 1 = 0$, k can be any element of $\mathbb{N}_n$. Thus x can be sent to any g . Thus the 1 dimensional representations of C_n are $\rho_0, \dots, \rho_{n-1}$ such that ρ_k takes $x \rightarrow \omega^k$.

Define $\rho_k^n(x) := \prod_{n=1}^m \rho_k(x)$. $\forall k \in \mathbb{N}_n$, $\forall x \in C_n$, $\rho_1^k(x) = \prod_{n=1}^m \omega = \omega^k = \rho_k$, moreover, $\rho_1^n(x) = \prod_{n=1}^n \omega = \omega^n = 1$. Thus $\widehat{G} \cong \mu_n = C_n$. $\square$

Problem 5.7.

Proof. Let $\widehat{\phi} : \widehat{G'} \rightarrow \widehat{G}$ which takes $\alpha \rightarrow \alpha \circ \phi$. $\alpha \in Hom_{gr}(G', \mathbb{C})$ and $\phi \in Hom_{gr}(G, G')$, thus $\alpha \circ \phi \in Hom_{gr}(G, \mathbb{C})$. Thus $\widehat{\phi}$ is a homomorphism from $\widehat{G'}$ to $\widehat{G}$

$\ker \widehat{\phi} = \{f : G' \rightarrow \mathbb{C}^* : f \circ \phi = 1\}$. $\forall f \in \ker \widehat{\phi}$, $\phi(G) \leq \ker f$, thus f factors through $G'/\phi(G)$. Thus there is a 1 to 1 correspondence of $\ker \widehat{\phi}$ and $Hom(G'/\phi(G), \mathbb{C})$, thus $\ker \widehat{\phi} \cong Hom(G'/\phi(G), \mathbb{C}) = G'/\phi(G)$. By 5.5, $|\ker \widehat{\phi}| = |\widehat{G'}/\widehat{\phi(G)}| = |\widehat{G'}|/|\widehat{\phi(G)}| = |G'|/|\phi(G)|$, by 5.5. Furthermore, $|\widehat{\phi(G)}| = |\widehat{G'}|/|\ker \widehat{\phi}|$ thus $|\ker \widehat{\phi}| = |\widehat{G'}|/|\widehat{\phi(G)}|$. Thus, $|\widehat{G'}|/|\widehat{\phi(G)}| = |G'|/|\phi(G)|$

$\implies$: Suppose ϕ is injective. Then $|G| = |\phi(G)|$. Thus, $|G'|/|\phi(G)| = |G'|/|G| = |G'|/|\widehat{G}|$, by 5.5.
 Thus $|G'|/|\widehat{\phi(G')}| = |G'|/|\widehat{G}| \implies |\widehat{\phi(G')}| = |\widehat{G}|$. Thus $\widehat{\phi(G')} \cong \widehat{G}$. Thus $\widehat{\phi}$ is surjective.

$\Leftarrow$: Suppose $\widehat{\phi}$ is surjective. Thus $|\widehat{\phi(G')}| = |\widehat{G'}| = |G'|$ by 5.5. Then $|G'|/|\widehat{\phi(G')}| = |G'|/|G| = |G'|/|\phi(G)| \implies |\phi(G)| = |G| \implies |\ker \phi| = 1$ because $|\phi(G)| = |G/\ker G| = |G|/|\ker G|$. Thus G is injective. $\square$